

# Picking an FPGA Board on Amazon

A buyer's method, a trap gallery, and the measured answer for the theremin

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## The one rule

**Marketing sells boards; part numbers decide them.** Every Amazon listing says “FPGA development board.” The only words that matter are the exact device marking on the chip — family, logic-cell count, speed grade, package — and you can almost always read it from the product photo. If a listing shows no part number and no legible chip photo, that is the answer: do not buy it.

## The five questions, in order

1. **Exactly which chip?** Find the marking (e.g. iCE40HX8K-CT256). Decode it: family (HX = high-performance, UP = ultra-low-power and *much slower*, LP = low-power), logic cells (1K = 1,280, 4K/8K = 7,680 — note the 4K and 8K are the same die, 5K = 5,280), package.
2. **Does my design fit — by measurement, not vibes?** You need three numbers from your own synthesis run: logic cells, block RAMs, and Fmax. Never size from the marketing number (“7,680 LUTs!”); run `yosys + nextpnr` for the exact part *before* buying. It costs nothing and takes minutes.
3. **How does it program?** Onboard USB (FTDI) means one data cable and `openFPGALoader`. No onboard programmer means a separate JTAG/SPI dongle and wiring — an invisible +\$30 and an evening of frustration on a board that looked cheap.
4. **Is the open toolchain supported?** For iCE40 (icestorm) and ECP5 (trellis) the answer is yes and the flow on this laptop works. For families like Gowin (Tang Nano) or old Altera parts sold as “FPGA boards,” you inherit a different or dead toolchain.
5. **Can I reach the pins I need?** Count the I/O the listing actually breaks out (PMOD headers, 0.1-inch holes) against what the design needs — the theremin needs two oscillator inputs, an audio pin, and a clock.

## Trap gallery: iCE40 listings that look right and are not

*Link honesty: these are Amazon search links, not verified product pages — I cannot browse, and a specific ASIN would be a guess. The board names are real and commonly sold; the disqualifying facts below are measured on this project's own design (**theremin\_top**: 2,432 LC / 6 EBR needed, 48 MHz target).*

| Trap                                                                                | Why it looks right                                          | Why it fails (measured)                                                                                                                             |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Lattice iCEstick</b><br>(iCE40HX1K) search                                       | Official Lattice eval kit, ~\$40,<br>“iCE40 FPGA USB stick” | HX1K = 1,280 LC. The theremin needs 2,432: <b>190% utilization, will not place.</b> Measured, not estimated.                                        |
| <b>Nandland Go Board</b><br>(iCE40HX1K) search                                      | Beginner-friendly, great tutorials, quality board           | Same 1,280-LC die. Excellent for lessons 0–8; cannot hold this design.                                                                              |
| <b>Olimex iCE40HX1K-EVB</b><br>search                                               | Cheap, open hardware                                        | HX1K again — and no onboard programmer (question 3).                                                                                                |
| <b>UPduino v3.x</b> (iCE40UP5K)<br>search                                           | “5,280 LUTs, biggest low-power iCE40,” ~\$25                | Fits at 46% LC — but UP-family silicon is slow: <b>measured Fmax 21.65 MHz</b> vs the 48 MHz plan. Works only if you re-plan the clock (see below). |
| <b>iCEBreaker / iCESugar</b><br>(iCE40UP5K) search                                  | Best-documented small FPGA boards around                    | Same UP5K story: capacity yes, speed no at 48 MHz.                                                                                                  |
| <b>Generic “iCE40 FPGA development board”</b> search                                | Cheapest listing on the page                                | No part number, no schematic, unknown programmer. Question 1 fails: unknowable, so no.                                                              |
| <b>“FPGA” boards that are not iCE40 at all</b> (Tang Nano, EP2C5 Cyclone II) search | Same shelf, same price, say “FPGA”                          | Different family: different (Gowin) or dead (Quartus 13-era Altera) toolchain. The flow you have proven does not apply.                             |

The pattern: **every trap is a true statement about the wrong requirement.** “Official Lattice” is true (wrong size). “5,280 LUTs” is true (wrong speed). “Cheap” is true (no programmer). The part number plus your own measured numbers cut through all of it.

## The nuance the UP5K teaches

A board that fails your stated constraint may still meet your actual requirement. The UP5K fails 48 MHz — but at its real 21 MHz the theremin’s pitch resolution is 0.20 cent, still ~25x below audibility. The lesson: know which constraints are physics (resolution must be inaudible) and which are habits (48 MHz was a convenient number). Cheap boards are sometimes fine; the measurement tells you when.

## The best choice for the theremin

**Alchitry Cu V2** (search) — an iCE40HX8K with onboard USB programming.

Measured on the exact design (HX8K/cb132): 2,432/7,680 LC (31%), 6/32 EBR, Fmax 55.6 MHz — passes the 48 MHz plan with timing margin and 3x logic headroom, programs over one USB

data cable, and the flow (`make pnr-ice40 ICE_DEV=hx8k ICE_PKG=cb132`) already produces its numbers. Before ordering, apply question 1 to the listing: confirm the chip photo reads iCE40HX8K. (An HX8K breakout board is on the BOM as owned but could not be located; if it turns up, it measures the same and costs nothing.)

## What this does *not* answer

The theremin is the de-risking vehicle. **MAIDEN's radar chain does not fit any iCE40:** measured, it needs 46 hard multipliers and ~8,200 LUT4 — HX-family iCE40 has zero multipliers. That decision (ECP5-class, ~\$315) was deliberately deferred, not solved, when the ULX3S order was canceled; it comes back at maiden36 with these same methods.